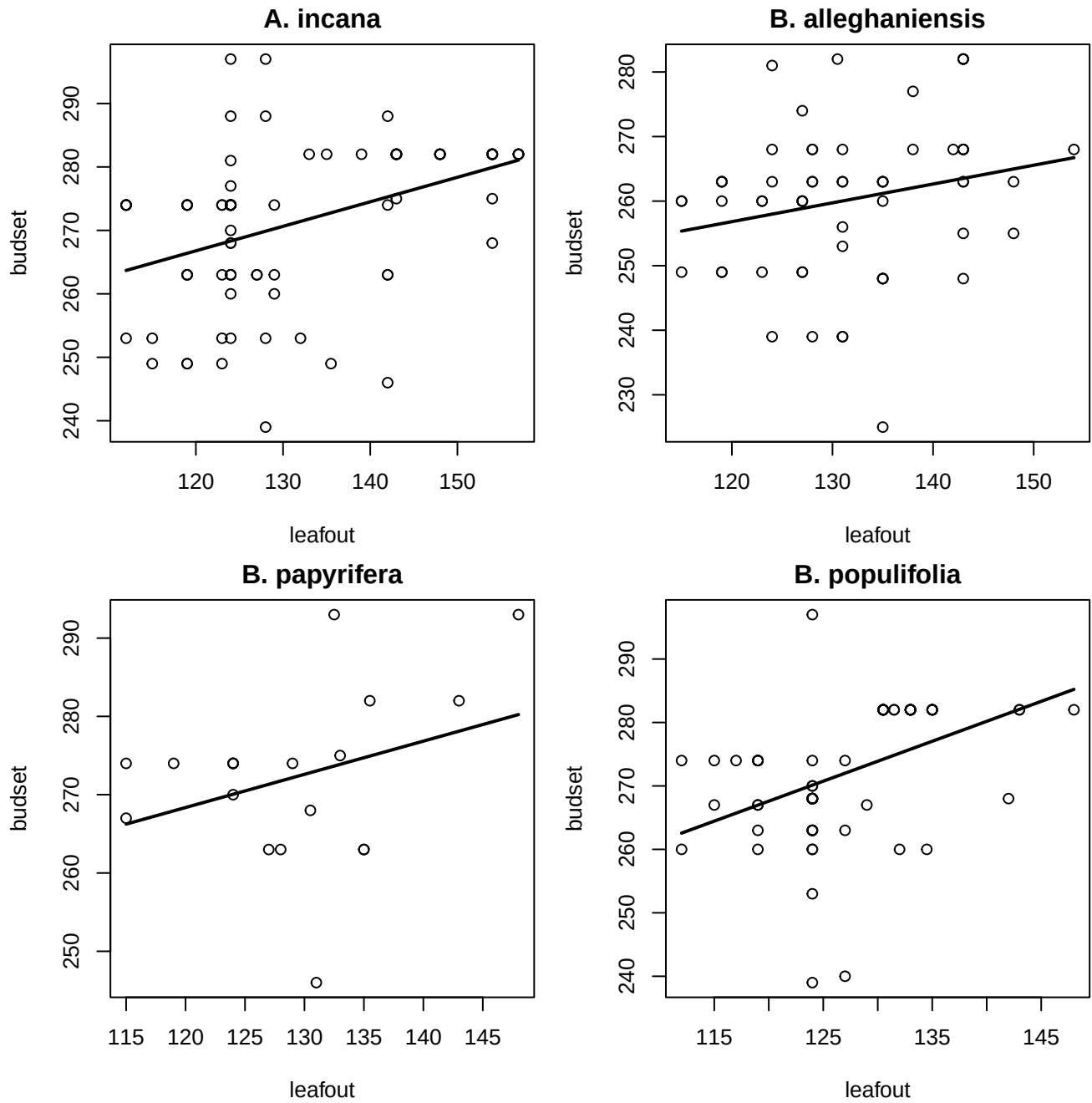


Results outline

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Research questions

1. Does tree growth increase with longer growing seasons?

2. Did an earlier start and later end of season increase growth?
3. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?
4. How did growth change across years, and how did it relate to climate?

Results

1. Summary basic information

- (a) Wildchokkie includes 81 individuals of four deciduous tree species from four sites with leaf phenology data spanning three years (2018 to 2020) with 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS).
- (b) GSL ranged from 90 days (*B. alleghaniensis* in 2018) to 173 days (*A. incana* in 2018)
- (c) GS thermal sum ranged from 1465.1 GDD (*B. alleghaniensis* in 2018) to 2513.6 GDD, (*A. incana* in 2018)
- (d) The earliest leafout day of year was 112 (2019) and the latest, 157 (2019). The earliest budset day of year was 225 (2018) and the latest, 297 (2018)

2. Does tree growth increase with longer growing seasons?

- (a) Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it.
- (b) Longer thermal season increased growth for *B. alleghaniensis* ($\beta = 0.13$, UI_{90} [0.01, 0.25]), *B. papyrifera* ($\beta = 0.44$, UI_{90} [0.17, 0.71]), and *B. populifolia* ($\beta = 0.21$, UI_{90} [0.07, 0.35]) (Figure 1a (Table S1)).
- (c) Calendar season had a weaker effect on growth than the thermal season across all species S4), and only two species grew more with longer calendar seasons (*A. incana*: $\beta = 0.05$, UI_{90} [0, 0.1], and *B. papyrifera*: $\beta = 0.18$, UI_{90} [0.06, 0.32]) (Figure 1b (Table S1)).
- (d) *B. alleghaniensis* and *B. populifolia* grew more with warmer thermal season, but didn't with longer calendar season

3. Did an later start and end of season increase growth?

- (a) A longer calendar season increased growth for two species, but a later start of season increased growth only for one of those two (*B. alleghaniensis*: $\beta = 0.1$, UI_{90} [0, 0.2], and decreased for (*A. incana*: $\beta = -0.07$, UI_{90} [-0.14, -0.01]) (Figure 1c (Table S1)).
- (b) Three species grew more with later end of season (*B. alleghaniensis*: $\beta = 0.12$, UI_{90} [0.05, 0.18], (*B. populifolia*: $\beta = 0.1$, UI_{90} [0.03, 0.16], with *B. papyrifera* (*B. papyrifera*: $\beta = 0.21$, UI_{90} [0.09, 0.33]) being the only one of those three that also grew more with longer calendar season (Figure 1d (Table S1)).

4. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?

- (a) *B. papyrifera* grew the least ($\beta = -3.76$, UI_{90} [-8.62, 1.12]) and *B. populifolia* grew the most ($\beta = -0.66$, UI_{90} [-4.86, 3.65]) (Figure S3).
- (b) Growth did not change across any site (Figure 2) (Table S2), but there was more variation across sites ($\sigma = 0.18$, UI_{90} [0.02, 0.5]) than across treeids ($\sigma = 0.17$, UI_{90} [0.05, 0.27])

5. How did growth change across years, and how did it relate to climate?

- (a) Growth was highest in 2018 (0.18, UI_{90} [-0.74, 1.12]), but it was neither the wettest, the driest, the shortest, the latest SOS, the earliest EOS nor the highest thermal sum, unlike our finding that warmer and longer seasons increase growth for most of our species (Figure 1 a and b).
- (b) Growth was lowest in 2020 (-0.44, UI_{90} [-1.36, 0.5]) and that year was the driest (also the only year with a drought that ramped up to "severe" in September) the shortest, the latest SOS and EOS and had the highest thermal sum, also unlike our findings that growth increases with warmer seasons.

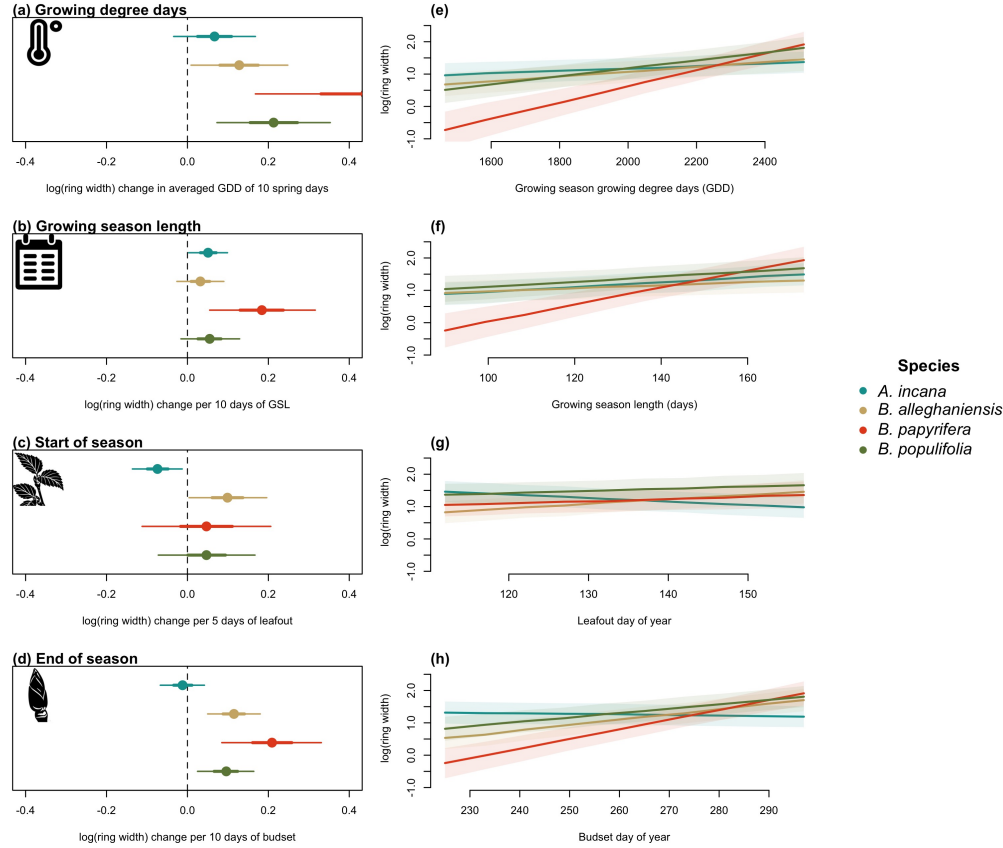


Figure 1: Model slope parameters across the different model predictors for Wildchrokie species. GDD represent the accumulated growing degree days between leafout and budset (a) the models estimates shows log ring width change per 10 average spring days GDD (174.05). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). All predictors were fit using the same model and number of observations, with only the independent variable changing. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI. *A. incana* and *B. papyrifera* grew more with longer calendar seasons, but only *A. incana* grew more with earlier leafout, but not with later budset. Interestingly, *B. papyrifera* did not grow more with earlier leafout, but did with a later budset. *B. alleghaniensis* and *B. populifolia* did not grow more with longer calendar seasons, but *B. alleghaniensis* grew more with a later leafout and budset whereas *B. populifolia* only grew more with a later budset.

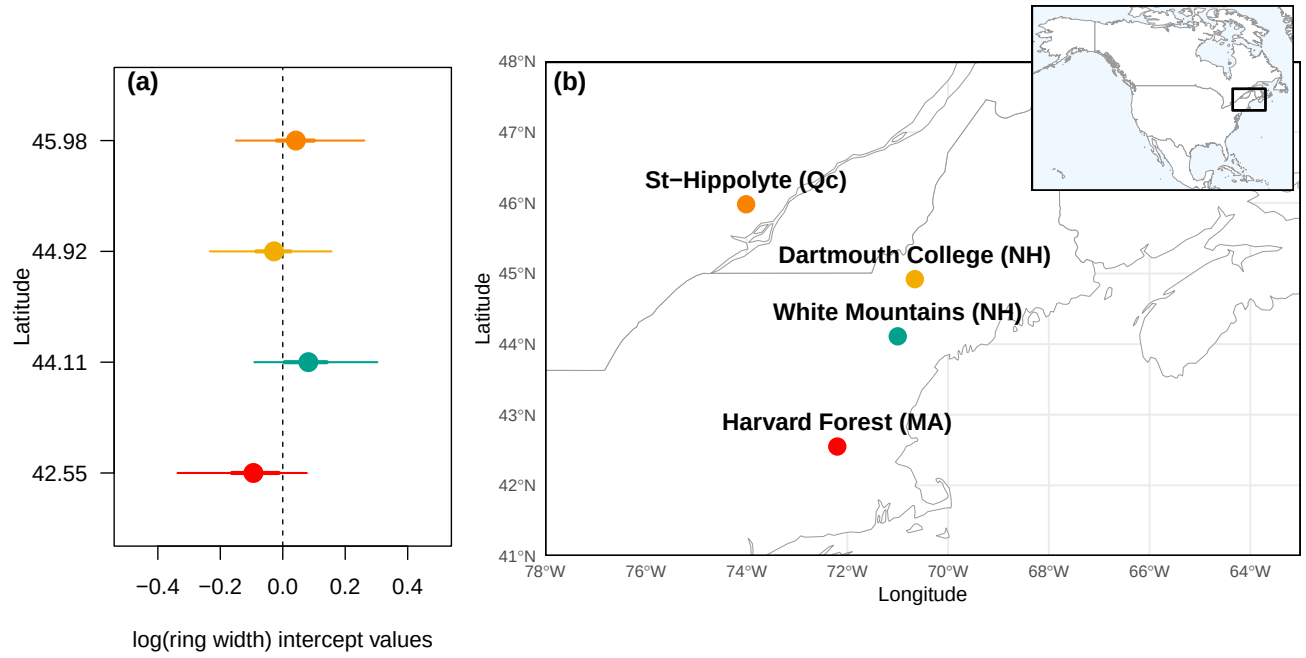


Figure 2: The effect of sites on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI. The sites are color-coded from (b), the map showing their locations.

Supplemental results

Table S1: Species level slope parameters (β and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
<i>A. incana</i>	$\beta = 0.07$, [-0.03, 0.17]	$\beta = 0.05$, [0.00, 0.10]	$\beta = -0.07$, [-0.14, -0.01]	$\beta = -0.01$, [-0.07, 0.04]
<i>B. alleghaniensis</i>	$\beta = 0.13$, [0.01, 0.25]	$\beta = 0.03$, [-0.03, 0.09]	$\beta = 0.10$, [0.00, 0.20]	$\beta = 0.12$, [0.05, 0.18]
<i>B. papyrifera</i>	$\beta = 0.44$, [0.17, 0.71]	$\beta = 0.18$, [0.06, 0.32]	$\beta = 0.05$, [-0.11, 0.21]	$\beta = 0.21$, [0.09, 0.33]
<i>B. populifolia</i>	$\beta = 0.21$, [0.07, 0.35]	$\beta = 0.06$, [-0.02, 0.13]	$\beta = 0.05$, [-0.07, 0.17]	$\beta = 0.10$, [0.03, 0.16]

Table S2: Site level intercept parameters (α and 90% uncertainty intervals, in square brackets) for the GDD model.

Site	Intercept.estimate	Latitude	Longitude
	$\alpha = -0.03$, [-0.23, 0.16]	44.79	-71.15
	$\alpha = -0.09$, [-0.34, 0.08]	42.53	-72.19
	$\alpha = 0.04$, [-0.15, 0.26]	45.93	-74.03
	$\alpha = 0.08$, [-0.09, 0.3]	44.11	-71.23

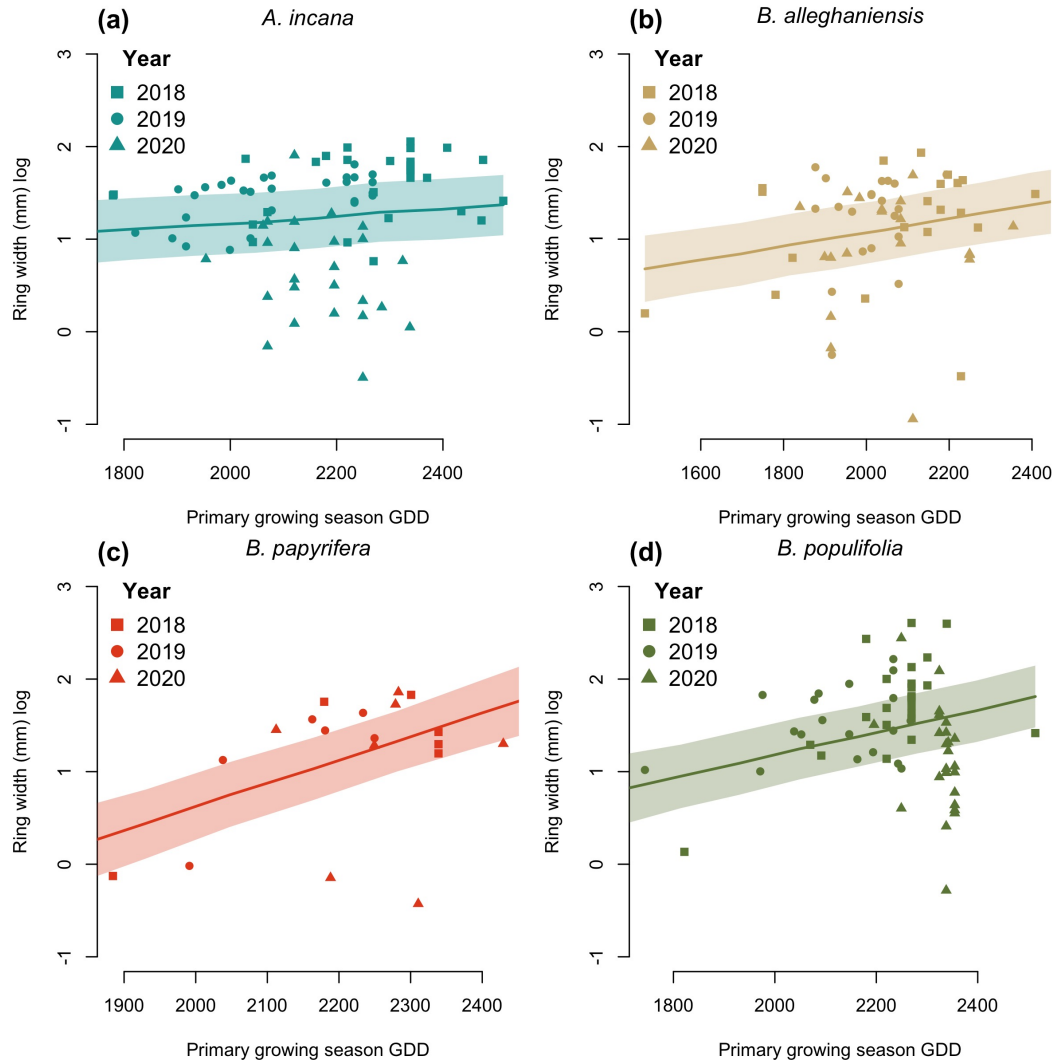


Figure S1: atreeid

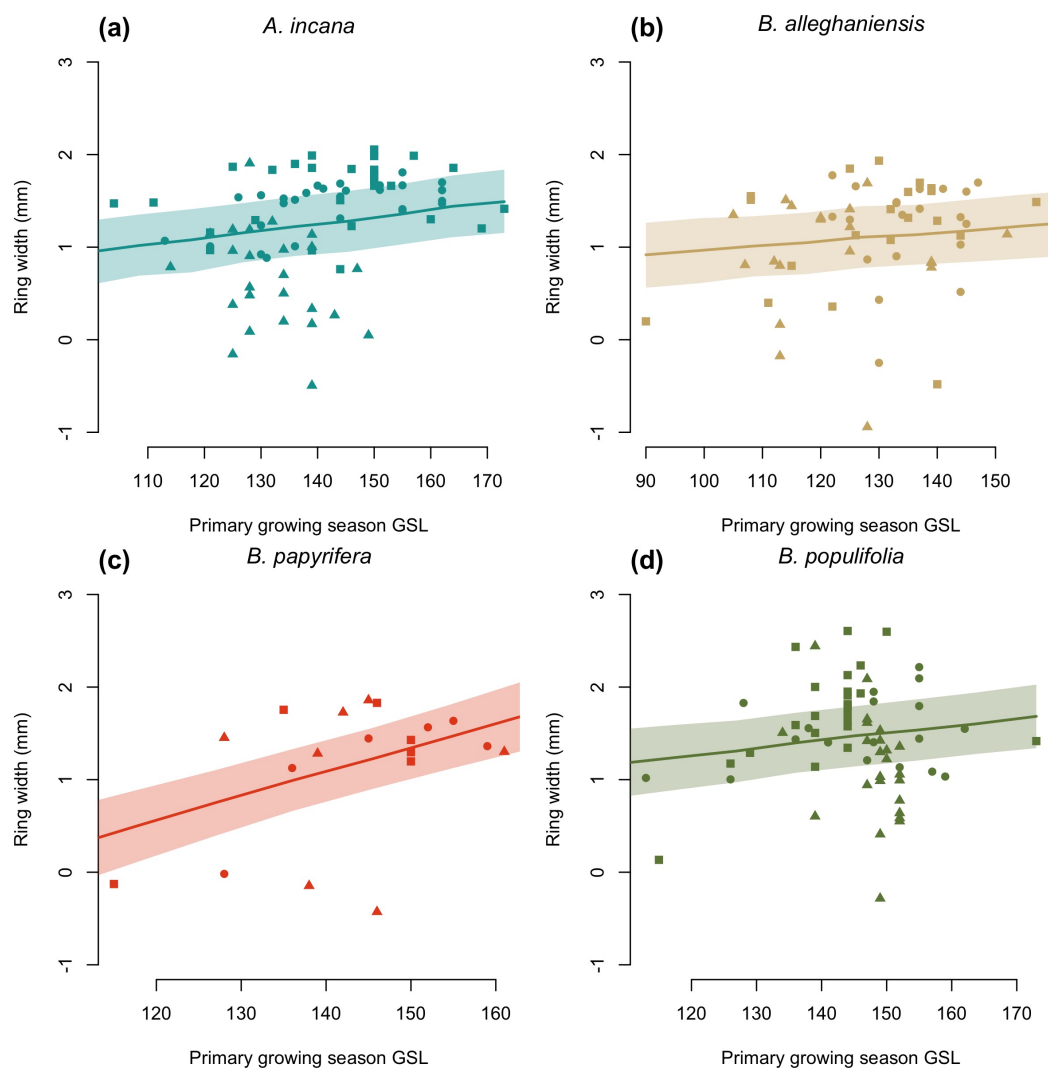
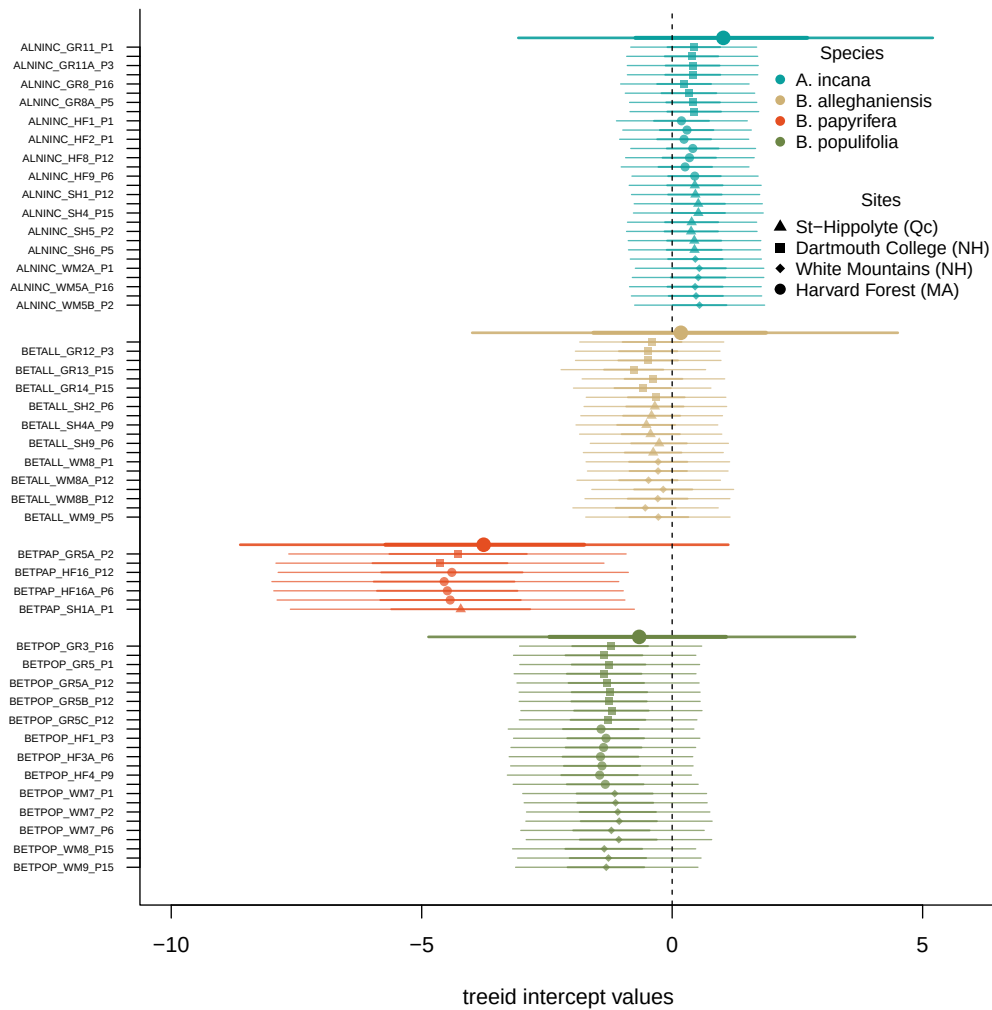


Figure S2: atreeid



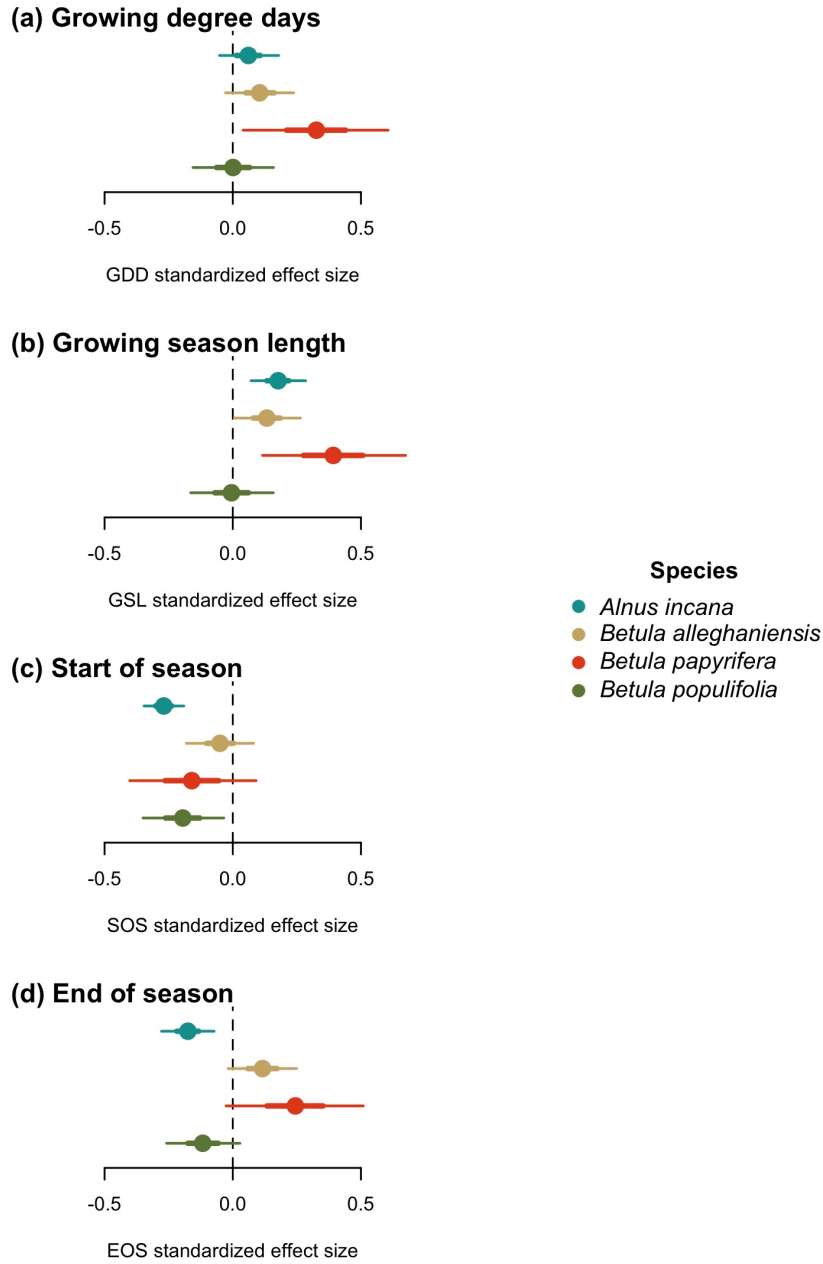


Figure S4: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.